

Pablo Leyva

908-525-6062 | pl33@njit.edu | linkedin.com/in/pablo-leyva | github.com/pleyva2004 | pabloleyva.io

EDUCATION

New Jersey Institute of Technology

Newark, NJ

B.S. in Applied Mathematics (Statistics) & Computer Science (AI: Learning and Reasoning)

Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Database Systems, Intro to AI, Intro to Machine Learning, Neurosymbolic AI, Explainable AI, Deep Learning, Reinforcement Learning, Probability, Linear Algebra, Convex Optimization

EXPERIENCE

Algorithms and Architectures for Reasoning and Intelligent Automation (ARIA) Lab

Sep 2026 – Present

Research Fellow

Newark, NJ

- Studying **semantic faithfulness at the neural-symbolic interface**: LLM-generated first-order logic that is solver-valid yet encodes a different proposition, so a theorem prover returns apparently rigorous reasoning over a wrong representation
- Building a **structure-aware semantic auditor** between formalizer and solver that decomposes and aligns the English and the NL→FOL output, emitting a **component-level diagnosis** (error category, corrupted span, expected vs. generated) instead of a confidence score, so flagged formalizations can be repaired rather than only rejected

Apple

May 2026 – August 2026

AI/ML Intern

Cupertino, CA

- Built **model-based tools** that curate and score **RL post-training data** for a continually-learning SLM; own the team's data-quality workstream studying how curation drives out-of-distribution generalization on agentic Apple Wallet and Apple Pay workflows
- Designing **RL environments with verifiable rewards** (program-analysis-style oracles for payment-flow correctness); implementing **GRPO/PPO** on-policy training with vLLM rollouts and a reward model trained on preference data

Apple

May 2025 – August 2025

AI/ML Intern

Cupertino, CA

- Built an **agent orchestration system** integrating the Apple Foundation Model (AFM) with Model Context Protocol (MCP); designed stateful multi-step tool execution with retry and human-in-the-loop fallback paths, collecting **trajectory-level preference data** across 10k+ rollouts
- Built the team's **LLM evaluation pipeline** benchmarking agent trajectories (tool-use success, groundedness, hallucination rate) against Claude and GPT baselines; results directly informed fine-tuning priorities
- Designed **tool selection and execution** flows with LangGraph using TouchID human-in-the-loop validation; presented the prototype at the team's end-of-internship review (org director present)

Undergrad Research and Innovation (URI) - Agentic Reasoning SLM

Sep 2025 – May 2026

Research Fellow

Newark, NJ

- Training a **3B-parameter reasoning SLM** via **SFT** → **DPO** → **RL fine-tuning** with multi-GPU distributed training (FSDP) on math-proof and logic-puzzle datasets, studying one-step reasoning generalization in small language models
- Constructed a **synthetic RL environment** with programmatically-checkable rewards for mathematical proof search; ran **ablation studies** across reward-shaping strategies and data-mixing ratios to isolate sources of reasoning gain
- Ran **linear probes** on hidden activations and **safety evaluations** (refusal consistency, reward hacking, out-of-distribution robustness), benchmarking the SLM against larger LLM baselines on lm-eval-harness

Caterpillar Inc.

May 2024 – August 2024

Software Engineering Intern

Chicago, IL

- Built an **end-to-end data pipeline** with Python and Apache Airflow (scheduled DAGs with retry and failure handling) ingesting 500K+ daily records; served model predictions via **FastAPI microservices** at sub-100ms latency for 10K+ daily requests
- Fine-tuned **transformer LLMs** (BERT, GPT-3.5) with **LoRA/QLoRA** on a 500K-record corpus; ran ablation studies on data mixing and learning-rate schedules, reaching 89% classification accuracy

PROJECTS

Mechanistic Interpretability of Reward Models | *PyTorch, TransformerLens, sae-lens, HuggingFace, Anthropic hh-rlhf* | **Python**

- Trained **sparse autoencoders** on mid-layer activations of open-source reward models (Skywork-Reward, DeBERTa-RM) over **preference-pair datasets** (hh-rlhf, UltraFeedback) to extract interpretable features driving the scalar reward
- Ranked SAE features by **causal influence on reward score** and flagged candidate **reward-hacking proxies** (response length, hedging, markdown formatting, sycophantic apology) that correlate with RM score but not with human preference
- Implemented **activation steering** mitigation that ablates hacky feature directions at inference, recovering ~8 pts of preference accuracy on held-out adversarial pairs; writeup in draft for Alignment Forum / arXiv

Self-Aligning Agent Harness | *PyTorch, TRL, Gymnasium, vLLM, LoRA, Weights & Biases* | **Python**

- Built an **agent harness that aligns its own policy**: the model **self-critiques and revises** on-policy trajectories against a natural-language constitution, mines **self-generated preference pairs (RLAIF)** from the critiques, and closes the loop with **iterative DPO on a LoRA adapter**
- Added **rejection-sampling fine-tuning** over vLLM rollouts filtered by programmatic verifiers, plus a **self-reward model** calibrated against held-out human judgments; ran ablations on judge strength and preference volume, tracking **reward hacking and refusal-consistency drift** across alignment rounds in W&B

LEADERSHIP

President, Vice President - External Affairs, Event Coordinator
Society of Hispanic Professional Engineers (SHPE)

October 2023 – Present
Newark, NJ

- Leading a team of 20 to organize professional events for 300+ students; grew external company partnerships 6x and averaged 200% growth in weekly member engagement
- Built and shipped the chapter's first **iOS app** in 27 days with a team of 5 engineers, growing it to **200+ users** in its first month

TECHNICAL SKILLS

Languages : Python, C++, JavaScript/TypeScript, CUDA, Java

Systems & Infrastructure : distributed training (DeepSpeed, FSDP), high-throughput inference (vLLM, Ray), FastAPI microservices, Apache Airflow pipelines

ML Frameworks : PyTorch, scikit-learn, JAX, Hugging Face Transformers, TRL

Reinforcement Learning : PPO, GRPO, DPO, RLHF, reward modeling, Gymnasium, RL environment design, preference data

LLM Training & Evaluation : SFT, LoRA/QLoRA, fine-tuning, distillation, synthetic data, lm-eval-harness, safety evaluations, ablation studies